

Q10 Generate the 3-permutations of $\{1, 2, 3, 4, 5\}$.

Solution.

Generating the 3-combinations, we have:

123, 124, 125, 134, 135, 145, 234, 235, 245, 345.

Note that we generated 10 3-combinations, and $\binom{5}{3} = 10$.

Now, permutating through each of the 10 3-combinations, we have:

123	124	125	134	135	145	234	235	245	345
132	142	152	143	153	154	243	253	254	354
312	412	512	413	513	514	423	523	524	534
321	421	521	431	531	541	432	532	542	543
231	241	251	341	351	451	342	352	452	453
213	214	215	314	315	415	324	325	425	435

In total, we have $10 \times 6 = 60$ 3-permutations. (note that $P(5, 3) = 5 \times 4 \times 3 = 60$)

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